A PROJECT REPORT
ON

“Grievance Redressal Portal for Institute”

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF

DIPLOMA IN COMPUTER ENGINEERING

SUBMITTED TO

MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

SUBMITTED BY

SR NO	NAME OF STUDENT	ENROLLMENT	SHEAT NO.
1	KANADE ATHARV RAMESHWAR	2200940101	423914
2	HALNE ADITYA PRALHAD	23510240129	423931
3	PATHAN SHAHIDKHAN MUSAKHAN	23510240148	423943
4	SANGLE MAHARUDRA RAOSAHEB	23510240160	423953

GUIDED BY

(Mr. M. A. ZAHED)

GOVERNMENT POLYTECHNIC JINTUR,
DISTRICT PRABHANI STATE – MAHARASHTRA

Academic year: 2025-2026

GOVERNMENT POLYTECHNIC JINTUR

DISTRICT – PARBHANI

CERTIFICATE

This is to certify that the project report entitled "**Grievance Redressal Portal for Institute**" has been successfully completed by the students of **Sixth Semester Diploma in Computer Engineering.**

SR NO	NAME OF STUDENT	ENROLLMENT	SHEAT NO.
1	KANADE ATHARV RAMESHWAR	2200940101	423914
2	HALNE ADITYA PRALHAD	23510240129	423931
3	PATHAN SHAHIDKHAN MUSAKHAN	23510240148	423943
4	SANGLE MAHARUDRA RAOSAHEB	23510240160	423953

In partial fulfillment of the requirements for the award of the Diploma in **Computer Engineering.** This project work has been submitted to the Department of Computer Engineering, Government Polytechnic, Jintur and is a record of the work carried out by the student(s) during the **academic year 2025–26**, as per the prescribed curriculum. The project work has been found to be satisfactory and is approved for submission.

Mr. M. A. ZAHED
Project Guide

Mr. R. D. KOKARE
Head of Department

External Examiner

Dr. G. A. Kanade Sir
Principle

ACKNOWLEDGEMENT

We take this opportunity to express our sincere gratitude to all those who have supported and guided us throughout the completion of our project titled **"Grievance Redressal Portal for Institute."** We would like to express our heartfelt thanks to **Dr. Kanade Sir, Principal of Government Polytechnic, Jintur**, for providing us with the opportunity and necessary facilities to carry out this project work. We are deeply grateful to **Prof. R. D. Kokare, Head of the Department of Computer Engineering**, for their valuable support, encouragement, and guidance during the project. We would like to extend our sincere thanks to our **Project Guide, Prof. M. A. Zahed**, for their continuous guidance, motivation, and valuable suggestions throughout the development of this project. Their support helped us complete this work successfully. We are also thankful to all the faculty members of the Computer Engineering Department for their cooperation and assistance whenever required. Finally, we would like to thank our friends and family for their constant support and encouragement.

Sr. No.	Name of Students	Signature
1.	KANADE ATHARV RAMESHWAR	
2.	HALNE ADITYA PRALHAD	
3.	PATHAN SHAHIDKHAN MUSAKHAN	
4.	SANGLE MAHARUDRA RAOSAHEB	

INDEX TABLE OF CONTENT

SR. NO.	TITLES (CONTENT)	PAGE NO.
1	**INTRODUCTION**	**1**
	1.1 Background of the Project	1
	1.2 Problem Definition	1
	1.3 Aim of the Project	1
	1.4 Objectives	2
	1.5 Need of the System	2
	1.6 Overview of the Proposed System	2
	1.7 Advantages of the System	3
2	**LITERATURE REVIEW**	**4**
	2.1 Introduction	4
	2.2 Existing Systems	4
	2.3 Technologies Used in Existing Systems	5
	2.4 Research Findings	5
	2.5 Proposed System Improvements	5
	2.6 Comparative Analysis	5
	2.7 Conclusion	6
3	**SCOPE OF THE PROJECT**	**7**
	3.1 Introduction	7
	3.2 Scope of the System	7
	3.3 Functional Scope	7
	3.4 Application Scope	8
	3.5 Limitations of the System	8
	3.6 Assumptions	8
	3.7 Future Scope	8
	3.8 Conclusion	8
4	**SYSTEM ANALYSIS**	**9**
	4.1 Introduction	9
	4.2 Existing System Analysis	9
	4.3 Proposed System Analysis	9
	4.4 Functional Requirements	9
	4.5 Non-Functional Requirements	11
	4.6 Hardware Requirements	12
	4.7 Software Requirements	12
	4.8 Feasibility Study	12
	4.9 Conclusion	12
5	**SYSTEM DESIGN**	**13**
	5.1 Introduction	13
	5.2 System Architecture	13
	5.3 Use Case Diagram	13

	5.4 ER Diagram (Entity Relationship Diagram)	14
	5.5 Data Flow Diagram (DFD)	15
	5.6 Database Design	16
	5.7 User Interface Design	17
	5.8 Conclusion	17
6	**IMPLEMENTATION**	**18**
	6.1 Introduction	18
	6.2 Development Environment	18
	6.3 Modules Implementation	18
	6.4 Technologies Used	19
	6.5 Security Implementation	20
	6.6 Conclusion	20
7	**SYSTEM TESTING**	**21**
	7.1 Introduction	21
	7.2 Objectives of Testing	21
	7.3 Types of Testing Performed	21
	7.4 Test Case Design	22
	7.5 Error Handling	22
	7.6 Security Testing	23
	7.7 Performance Testing	23
	7.8 Conclusion	23
8	**RESULTS AND DISCUSSION**	**24**
	8.1 Introduction	24
	8.2 System Results	24
	8.3 Screenshots and Description	25
	8.4 Discussion	30
	8.5 Conclusion	30
9	**CONCLUSION AND FUTURE SCOPE**	**31**
	9.1 Conclusion	31
	9.2 Achievements of the System	31
	9.3 Limitations	31
	9.4 Future Scope	31
	9.5 Conclusion	32
	References	32
10	**APPENDICES**	**34**
	10.1 Introduction	34
	10.2 Appendix A: Sample Code	34
	10.3 Appendix A: Database Structure	35
	10.4 Appendix B: Test Data	35
	10.5 Conclusion	35
11	**SOURCE CODE LINK AND EXECUTION INSTRUCTIONS**	**36**
	11.1 Introduction	36
	11.2 GitHub Repository Link	36

11.3 Steps to Run the Project 36
12.4 Important Notes 37
12.5 Conclusion 37

LIST OF TABLES

Sr. No.	Title Of Table	Page No.
1	Comparative Analysis	6
2	Student Table database	16
3	Complaint Table database	17
4	Test Case Table	22
5	Database Structure	35

LIST OF DIAGRAMS

Sr. No.	Title Of Diagram	Page No.
1	Use Case Diagram	13
2	ER Diagram (Entity Relationship Diagram)	15
3	Data Flow Diagram (DFD)	16
4	Home Page	24
5	Registration Page	25
6	OTP Verification Page	25
7	Login Page	26
8	Student Dashboard	27
9	Complaint Submission Form	27
10	Complaint Tracking Page	27
11	Admin Dashboard	29
12	Admin Complaint Table	30

CHAPTER 1: INTRODUCTION

1.1 Background of the Project

In modern educational institutions, managing student grievances efficiently is a critical requirement. A grievance can be defined as any dissatisfaction or complaint raised by a student regarding academic, administrative, or infrastructural issues within the college. Traditionally, these grievances are handled manually through written applications or informal communication channels, which leads to inefficiency, lack of transparency, and delayed resolutions.

With the advancement of web technologies, there is a need to digitize such processes to ensure better communication, accountability, and faster resolution. The Grievance Redressal Portal is designed as a web-based system that enables students to register complaints and track their progress while allowing administrators to manage and resolve issues systematically.

1.2 Problem Definition

The existing manual grievance handling system suffers from several limitations:

- Complaints are recorded on paper, making them difficult to manage.
- No proper tracking mechanism for students.
- Lack of transparency in complaint resolution.
- Delayed responses due to manual processes.
- No centralized database for complaints.
- Difficulty in maintaining historical records.

Due to these issues, students often feel dissatisfied with the grievance handling process.

1.3 Aim of the Project

The primary aim of this project is to develop a secure, efficient, and user-friendly web-based grievance redressal system for college students.

1.4 Objectives

The main objectives of the system are:

- To provide an online platform for students to submit complaints.
- To ensure secure authentication using OTP verification.
- To enable students to track complaint status.
- To allow administrators to manage and resolve complaints.
- To maintain a structured database of all complaints.
- To provide role-based access control.

1.5 Need of the System

The need for this system arises due to:

- Increasing number of student grievances.
- Inefficiency of manual systems.
- Requirement for transparency and accountability.
- Need for quick and effective communication.

1.6 Overview of the Proposed System

The proposed system is a web-based portal consisting of two main users:

1. Student
2. Admin

<u>Student Features:</u>

- Registration with OTP verification
- Login and dashboard access
- Complaint submission
- Complaint tracking
- Password reset and account deletion

Admin Features:

- Login with fixed credentials
- View all complaints
- Update complaint status
- Filter and search complaints
- Delete old complaints

1.7 Advantages of the System

- Reduces manual work
- Improves transparency
- Provides real-time tracking
- Ensures data security
- Centralized data management

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Literature review involves studying existing systems and technologies related to grievance handling. It helps in understanding the limitations of current systems and identifying improvements for the proposed system.

2.2 Existing Systems

1. Manual Grievance System

In many colleges, grievances are handled manually through written applications.

Limitations:

- Time-consuming
- Difficult to manage records
- No tracking system
- Lack of transparency

2. Email-Based Systems

Some institutions use email for complaint submission.

Limitations:

- No structured format
- Difficult to categorize complaints
- No status tracking
- No centralized database

3. Basic Web Forms

Some systems provide simple online forms without advanced features.

Limitations:

- No authentication
- No complaint tracking
- Limited functionality

2.3 Technologies Used in Existing Systems

- HTML/CSS for frontend
- PHP/Java for backend
- MySQL databases
- Email systems for communication

2.4 Research Findings

From the study of existing systems, the following issues were identified:

- Lack of security mechanisms
- Poor user interface
- No real-time updates
- Limited administrative control

2.5 Proposed System Improvements

The proposed system addresses these issues by:

- Implementing OTP-based authentication
- Providing a user-friendly dashboard
- Enabling real-time complaint tracking
- Allowing admin control over complaint status
- Maintaining a structured database

2.6 Comparative Analysis

Feature	Manual System	Email System	Proposed System
Tracking	No	No	Yes
Security	Low	Medium	High
Speed	Slow	Medium	Fast
Transparency	Low	Low	High
Data Storage	Manual	Scattered	Centralized

Table No. 1

2.7 Conclusion

The literature review clearly indicates the need for a modern, secure, and efficient grievance management system. The proposed system overcomes the limitations of existing methods and provides a comprehensive solution.

CHAPTER 3: SCOPE OF THE PROJECT

3.1 Introduction

The scope of the project defines the boundaries and functionalities of the system. It outlines what the system will and will not do.

3.2 Scope of the System

The system is designed to:

Provide an online complaint submission platform

Enable secure user authentication

Allow complaint tracking

Provide admin control for complaint management

Maintain complaint history

3.3 Functional Scope

Student Side:

- Registration with OTP
- Login and dashboard
- Submit complaints
- Track complaints
- Reopen complaints
- Reset password
- Delete account

Admin Side

- Login with credentials
- View all complaints
- Filter and search complaints
- Update complaint status
- Delete old complaints

3.4 Application Scope

The system can be used in:

1. Colleges
2. Universities
3. Educational institutions

3.5 Limitations of the System

1. Requires internet connection
2. Limited to web platform (no mobile app)
3. Admin credentials are fixed
4. No multi-admin system

3.6 Assumptions

Users have basic knowledge of web applications

Email service is properly configured

Server environment is correctly set up

3.7 Future Scope

The system can be enhanced by:

- Developing a mobile application
- Adding AI chatbot support
- Implementing SMS notifications
- Adding analytics dashboard
- Integrating with college ERP system

3.8 Conclusion

The project scope ensures that the system meets the requirements of a college grievance handling process while leaving room for future enhancements.

CHAPTER 4: SYSTEM ANALYSIS

4.1 Introduction

System analysis is the process of studying and understanding the requirements of the system before development. It involves identifying user needs, system functionality, and constraints. This phase ensures that the developed system meets all user expectations.

4.2 Existing System Analysis

The existing grievance handling system in many colleges is either manual or partially digital.

Problems in Existing System:

- Manual complaint handling
- No centralized database
- No complaint tracking system
- Lack of transparency
- Delayed complaint resolution
- Poor communication between students and administration

4.3 Proposed System Analysis

The proposed system is a web-based application that provides:

- Secure login using OTP verification
- Complaint submission and tracking
- Admin control for complaint management
- Real-time status updates

4.4 Functional Requirements

Functional requirements describe what the system should do.

<u>1. User Registration</u>

1. Students can register using their details

2. OTP is sent to email for verification
3. Account is created after OTP validation

2. User Login

1. Students log in using enrollment number and password
2. System validates credentials
3. Redirect to dashboard on success

3. Complaint Submission

1. Student fills complaint form
2. Select category, subject, description
3. Upload supporting document
4. Submit complaint

4. Complaint Tracking

View complaint summary:

1. Total complaints
2. Pending
3. In Progress
4. Resolved
5. Rejected
6. View detailed complaint list

5. Complaint Reopening

1. Student can reopen resolved complaint
2. Provide reason for reopening

6. Password Reset

1. Enter enrollment and password
2. OTP verification
3. Set new password

7. Account Deletion

1. Enter credentials
2. OTP verification
3. Delete account and complaints

8. Admin Functions

1. Login with fixed credentials
2. View all complaints
3. Filter complaints
4. Change complaint status
5. Delete old complaints

4.5 Non-Functional Requirements

1. Security

1. OTP verification
2. Password protection
3. Secure database access

2. Performance

1. Fast response time
2. Efficient data handling

3. Usability

1. Simple interface
2. Easy navigation

4. Reliability

1. Accurate data storage
2. Consistent performance

5. Scalability

1. Can handle increasing number of users

4.6 Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4GB
- Storage: 20GB available
- Internet connection

4.7 Software Requirements

- Operating System: Windows/macOS/Linux
- Frontend: HTML, CSS
- Backend: PHP
- Database: MySQL
- Server: XAMPP
- Mail Service: PHPMailer

4.8 Feasibility Study

1. Technical Feasibility: The system uses widely available technologies like PHP and MySQL, making it technically feasible.

2. Economic Feasibility: The system is cost-effective as it uses open-source tools.

3. Operational Feasibility: The system is easy to use and requires minimal training.

4.9 Conclusion

System analysis ensures that all functional and non-functional requirements are clearly defined, making development efficient and structured

CHAPTER 5: SYSTEM DESIGN

5.1 Introduction

System design defines the architecture and components of the system. It includes diagrams and database design that represent system structure and flow.

5.2 System Architecture

The system follows a 3-tier architecture:

1. Presentation Layer (Frontend)
2. Application Layer (Backend)
3. Database Layer

5.3 Use Case Diagram

Actors:

1. Student
2. Admin

Use Cases:

1. Register
2. Login
3. Submit Complaint
4. Track Complaint
5. Reopen Complaint
6. Manage Complaints (Admin)

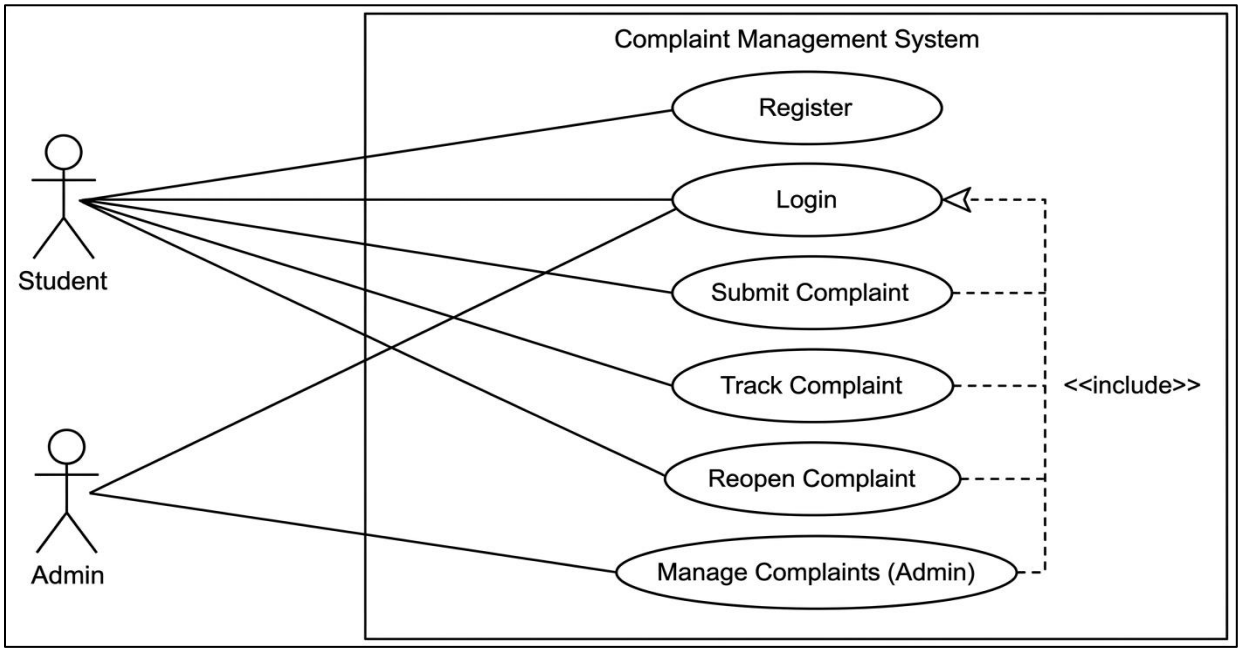


Diagram No. 1

5.4 ER Diagram (Entity Relationship Diagram)

Entities:

1. Student
2. Complaint
3. Admin

Attributes:

- Student:

1. student_id (Primary Key)
2. name
3. email
4. password

- Complaint:

1. complaint_id (Primary Key)
2. student_id (Foreign Key)
3. subject
4. category
5. description
6. status
7. attachment
8. submitted_on

- Admin:

1. admin_id
2. username
3. password

- Relationships:

1. One Student → Many Complaints
2. Admin manages Complaints

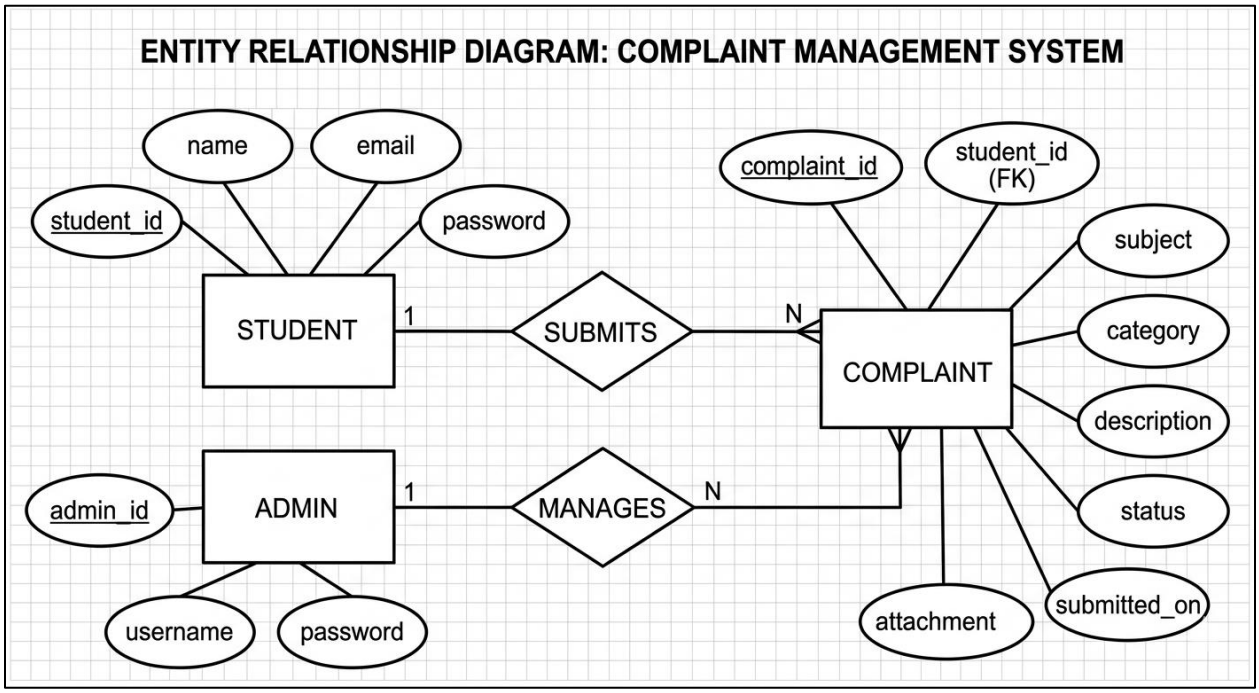


Diagram No. 2

5.5 Data Flow Diagram (DFD)

Level 0 (Context Diagram):

- External Entities:

1. Student
2. Admin
3. System: Grievance Portal

- Flow:

1. Student sends data → System
2. System processes → Admin

Level 1 DFD

- Processes:

1. Registration

2. Login
3. Complaint Submission
4. Complaint Tracking
5. Admin Management

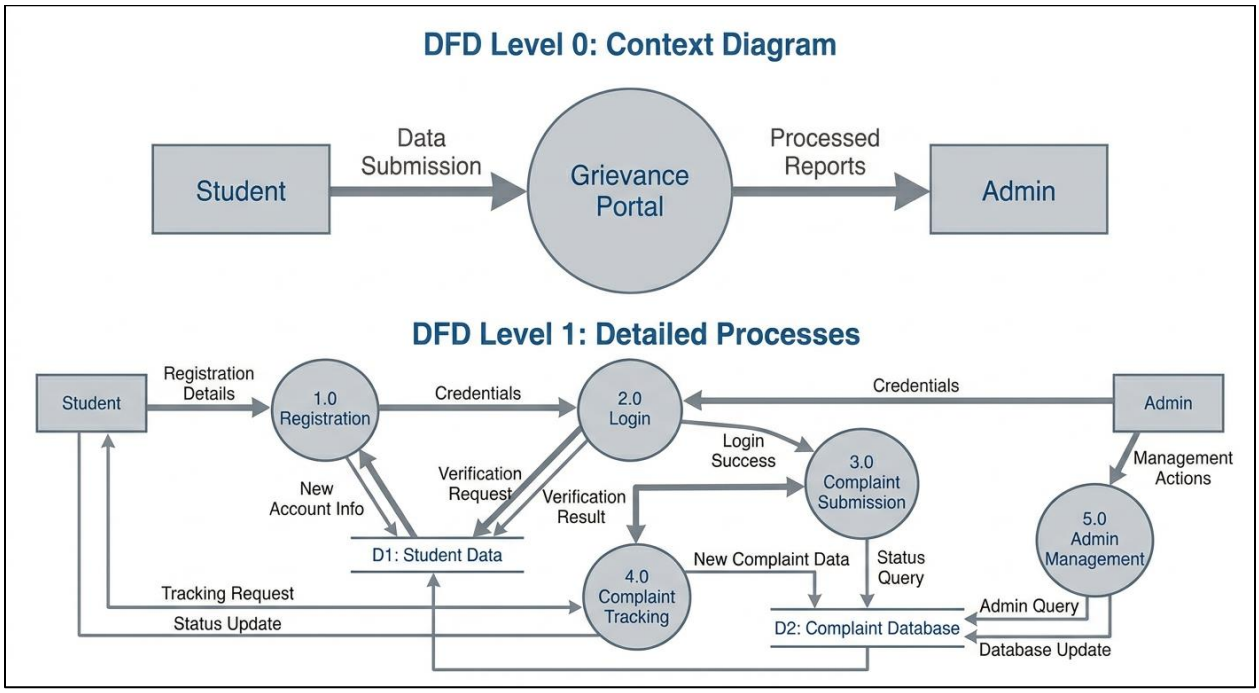


Diagram No. 3

5.6 Database Design

<u>Student Table:</u>

The student table stores core credentials including student_id, name, email, and encrypted passwords. It is essential for managing registration and secure authentication within the Grievance Redressal Portal.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	srno	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	name	varchar(30)	utf8mb4_general_ci		No	*None*			Change Drop More
3	student_id	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More
4	email	varchar(225)	utf8mb4_general_ci		No	*None*			Change Drop More
5	password	varchar(225)	utf8mb4_general_ci		No	*None*			Change Drop More
6	date	date			No	current_timestamp()			Change Drop More
7	course	varchar(30)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More

Table No. 2

Complaint Table:

The Complaint table logs student grievances using complaint_id (PK) and student_id (FK). It stores the subject, category, description, attachments, status, and submission date to track and manage the resolution process.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	**full_name**	varchar(100)	utf8mb4_general_ci		No	*None*			Change Drop More
3	**student_id**	varchar(50)	utf8mb4_general_ci		No	*None*			Change Drop More
4	**email**	varchar(100)	utf8mb4_general_ci		No	*None*			Change Drop More
5	**course**	varchar(50)	utf8mb4_general_ci		No	*None*			Change Drop More
6	**complaint_category**	varchar(100)	utf8mb4_general_ci		No	*None*			Change Drop More
7	**title**	varchar(255)	utf8mb4_general_ci		No	*None*			Change Drop More
8	**description**	text	utf8mb4_general_ci		No	*None*			Change Drop More
9	**status**	varchar(20)	utf8mb4_general_ci		Yes	Pending			Change Drop More
10	**submitted_at**	timestamp			No	current_timestamp()			Change Drop More
11	**attachment**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
12	**reopen_reason**	text	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
13	**reopened_at**	datetime			Yes	*NULL*			Change Drop More

Table No. 3

5.7 User Interface Design

- Student Interface

1. Dashboard
2. Complaint Form
3. Complaint List

- Admin Interface

1. Dashboard
2. Complaint Table
3. Filters and Actions

5.8 Conclusion

System design provides a blueprint for development, ensuring proper structure and organization.

CHAPTER 6: IMPLEMENTATION

6.1 Introduction

Implementation is the phase where the system is developed using selected technologies.

6.2 Development Environment

1. XAMPP server
2. PHP backend
3. MySQL database
4. HTML/CSS frontend

6.3 Modules Implementation

1. Registration Module

1. Collect student details
2. Generate OTP
3. Send OTP via email using PHPMailer
4. Verify OTP
5. Create account

2. Login Module

1. Validate enrollment and password
2. Start session
3. Redirect to dashboard

3. Dashboard Module

1. Display summary:
2. Total complaints
3. Pending

4. Resolved

4. Complaint Submission Module

Auto-fill student data (Input:)

1. Category
2. Subject
3. Description
4. Upload file
5. Save to database

5. Complaint Tracking Module

1. Display complaints in table format
2. Show status updates
3. Allow reopening

6. Admin Module

1. View all complaints
2. Filter by status
3. Update complaint status

7. OTP Module

1. Generate random OTP
2. Send email
3. Validate OTP

6.4 Technologies Used

Frontend

1. HTML: Structure
2. CSS: Design

Backend

1. PHP: Logic and processing

Database

1. MySQL: Data storage

Email Integration

1. PHPMailer: Sending OTP emails

6.5 Security Implementation

1. Password protection
2. OTP verification
3. Session management

6.6 Conclusion

Implementation ensures that all modules are properly developed and integrated to form a complete system.

CHAPTER 7: SYSTEM TESTING

7.1 Introduction

Testing is a crucial phase in the software development lifecycle. It ensures that the developed system works correctly, meets the requirements, and is free from errors. The Grievance Redressal Portal has been tested thoroughly to verify its functionality, performance, and security.

7.2 Objectives of Testing

- To verify system functionality
- To detect and fix errors
- To ensure system reliability
- To validate user requirements
- To check system performance

7.3 Types of Testing Performed

1. Unit Testing

- Each module was tested individually:
- Registration module
- Login module
- Complaint submission module
- Admin module

2. Integration Testing

All modules were integrated and tested together to ensure smooth interaction.

3. System Testing

Complete system testing was performed to verify overall functionality.

4. User Acceptance Testing (UAT)

The system was tested from the user's perspective to ensure usability and correctness.

7.4 Test Case Design

Test Case Table:

Test Case ID	Module	Test Description	Input	Expected Output	Actual Output	Status
TC01	Registration	Register new student	Valid details	Account created	Account created	Pass
TC02	OTP Verification	Verify OTP	Correct OTP	Verified	Verified	Pass
TC03	Login	Login with valid credentials	Correct ID & Password	Dashboard access	Dashboard shown	Pass
TC04	Complaint	Submit complaint	Valid data	Complaint saved	Saved successfully	Pass
TC05	Tracking	View complaints	Student login	List displayed	Displayed	Pass
TC06	Reopen	Reopen complaint	Reason given	Status updated	Updated	Pass
TC08	Admin	Change status	Select status	Status updated	Updated	Pass
TC09	Delete	Delete account	Valid OTP	Account deleted	Deleted	Pass
TC10	Login	Invalid login	Wrong password	Error message	Error shown	Pass

Table No. 4

7.5 Error Handling

The system handles errors such as:

1. Invalid login credentials
2. Incorrect OTP
3. Missing form fields
4. File upload errors

7.6 Security Testing

- OTP verification ensures user authenticity
- Password protection implemented
- Session management prevents unauthorized access

7.7 Performance Testing

- System responds quickly to user actions
- Handles multiple complaints efficiently

7.8 Conclusion

Testing confirms that the system is reliable, secure, and meets all functional requirements.

CHAPTER 8: RESULTS AND DISCUSSION

8.1 Introduction

This chapter presents the results obtained after implementing the system. It also includes screenshots and explanation of system modules.

8.2 System Results

The system successfully performs:

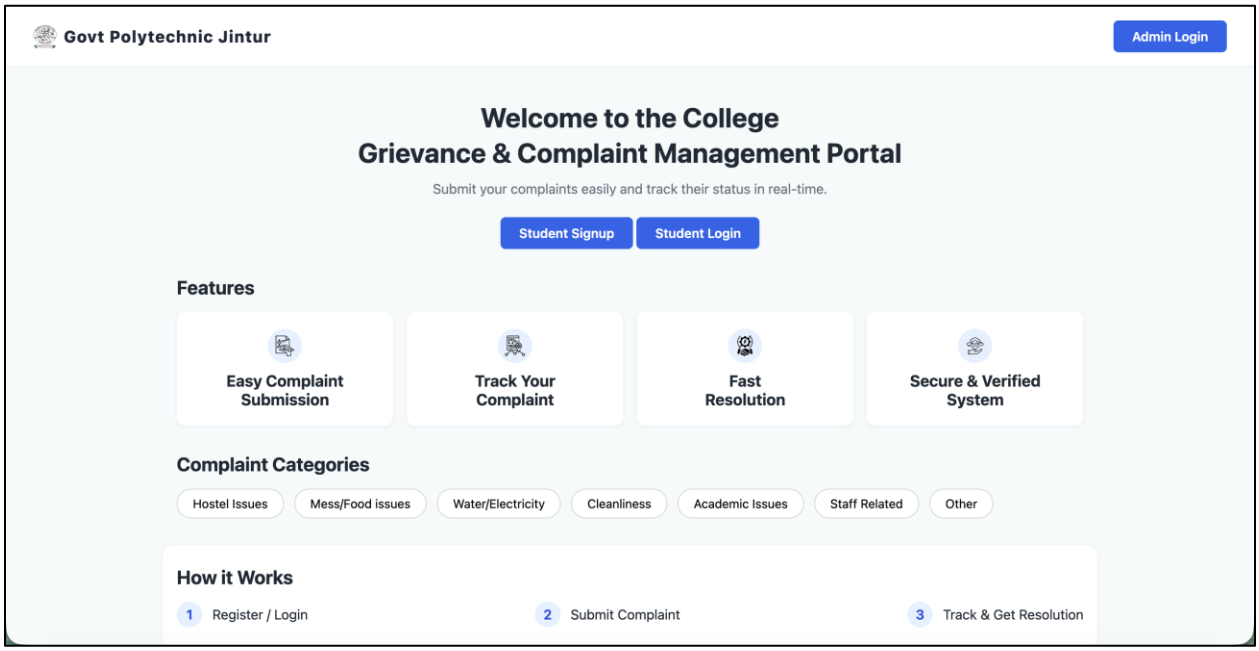
1. User registration with OTP
2. Secure login
3. Complaint submission
4. Complaint tracking
5. Admin management of complaints

8.3 Screenshots and Description

1. Home Page

Description:

- Displays system title
- Contains student options and admin login
- User-friendly interface

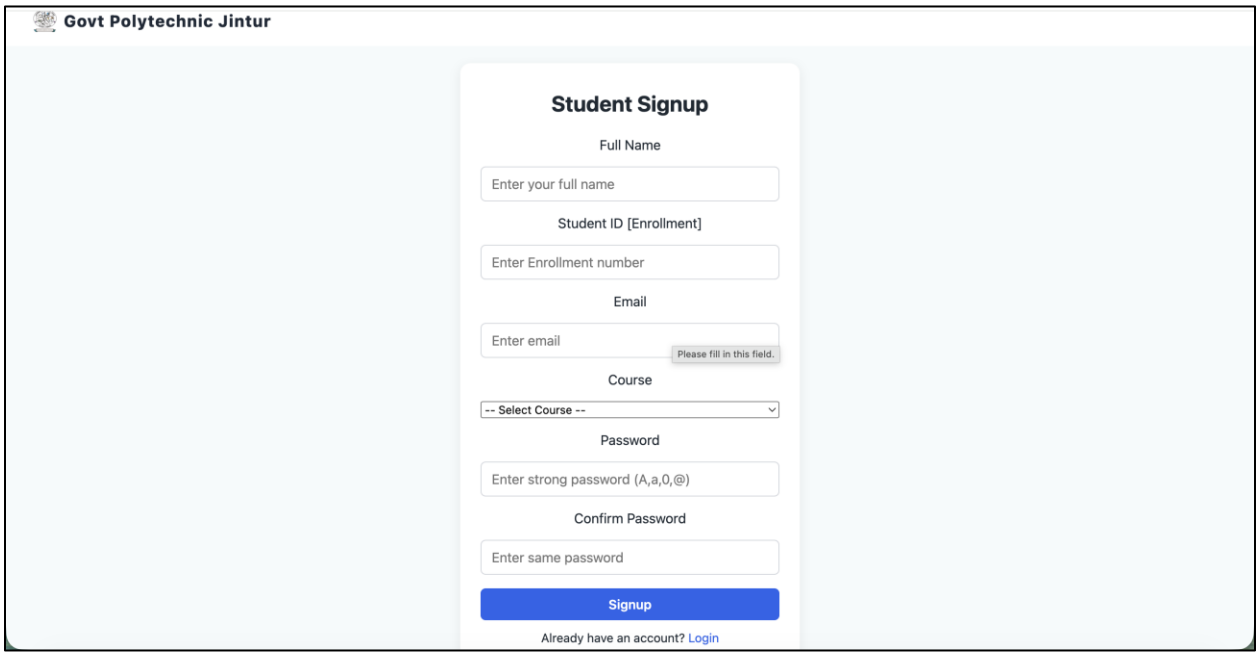


Screenshot No. 1

2. Registration Page

Description:

- Form for entering student details
- Includes email for OTP verification

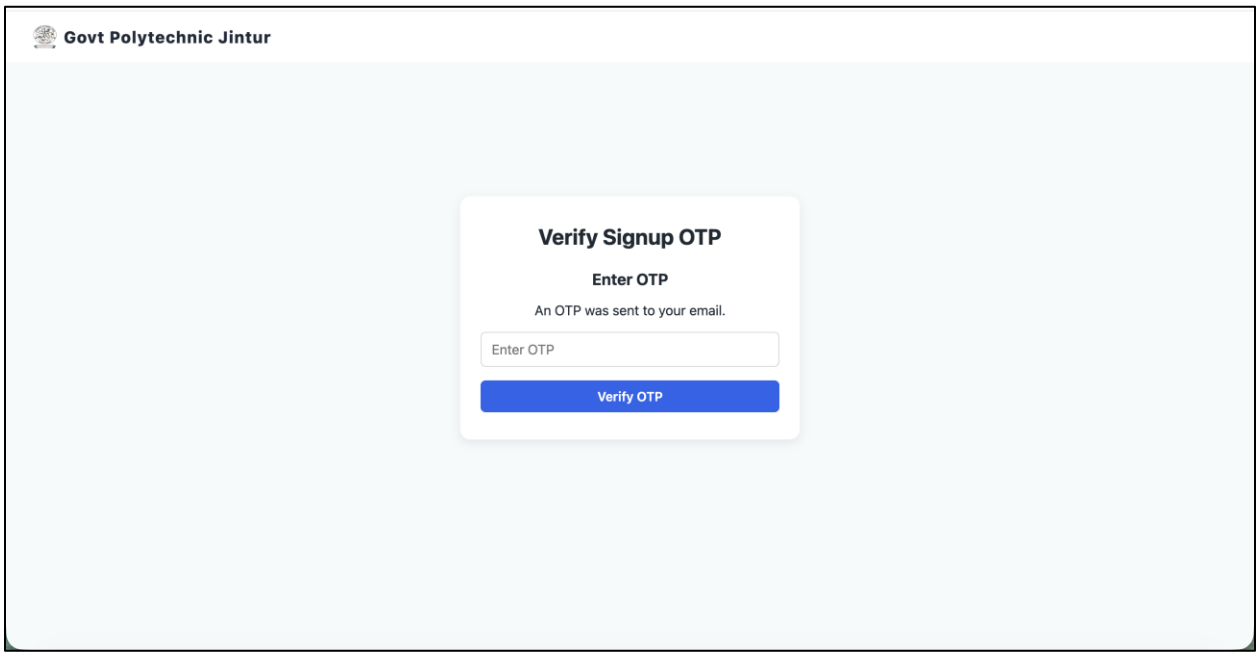


Screenshot No. 2

3. OTP Verification Page

Description:

- User enters OTP received via email
- Validates account creation

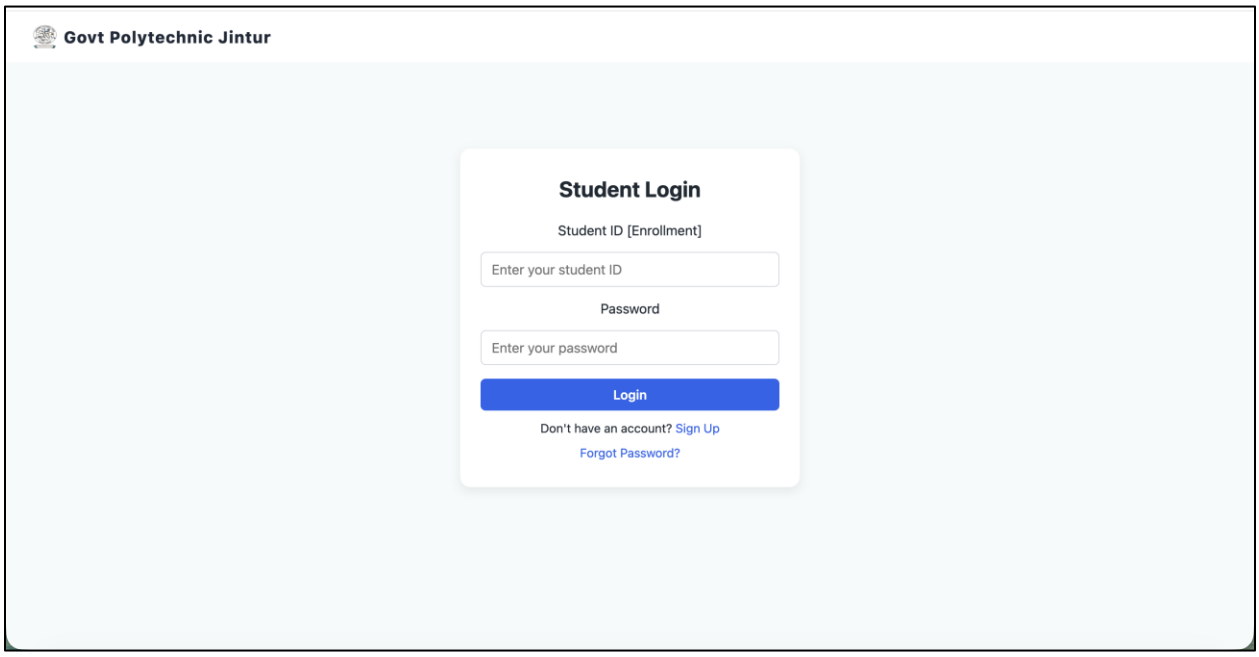


Screenshot No. 3

4. Login Page

Description:

- User enters enrollment and password
- Redirects to dashboard on success

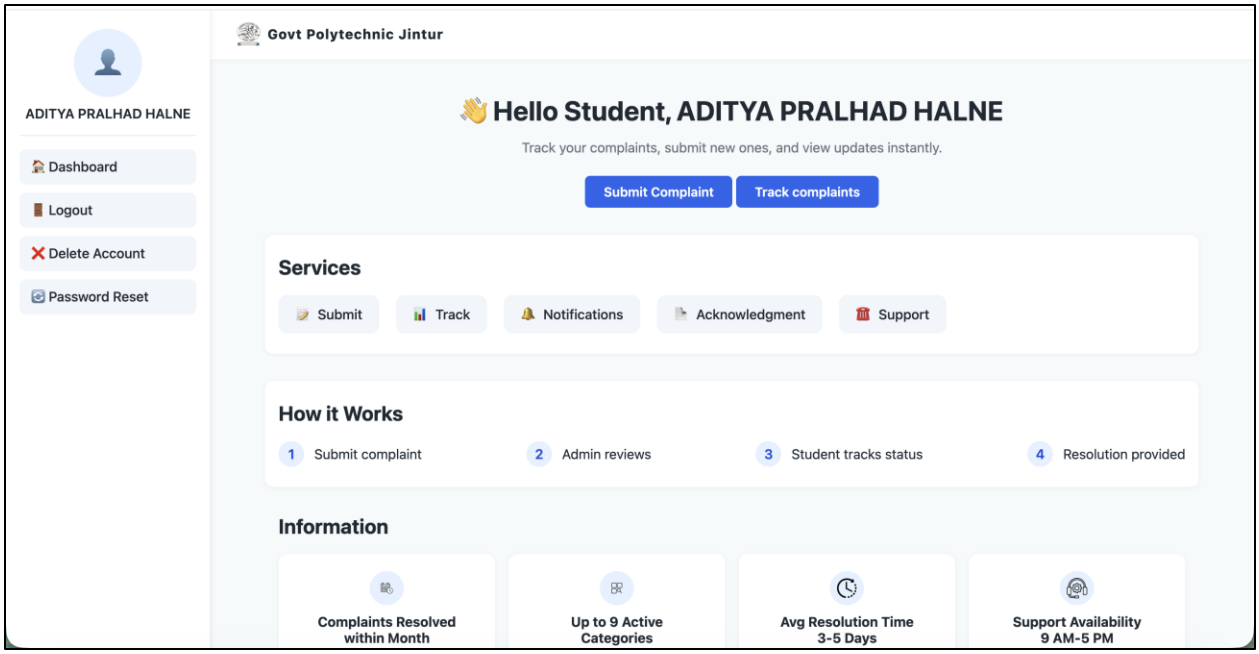


Screenshot No. 4

5. Student Dashboard

Description:

- Displays summary:
- Total complaints, Pending
- Resolved
- Navigation menu
- Student Data
- Menu items
- Logout
- Account deletion
- Password reset
- Submit complaint
- Track complaint

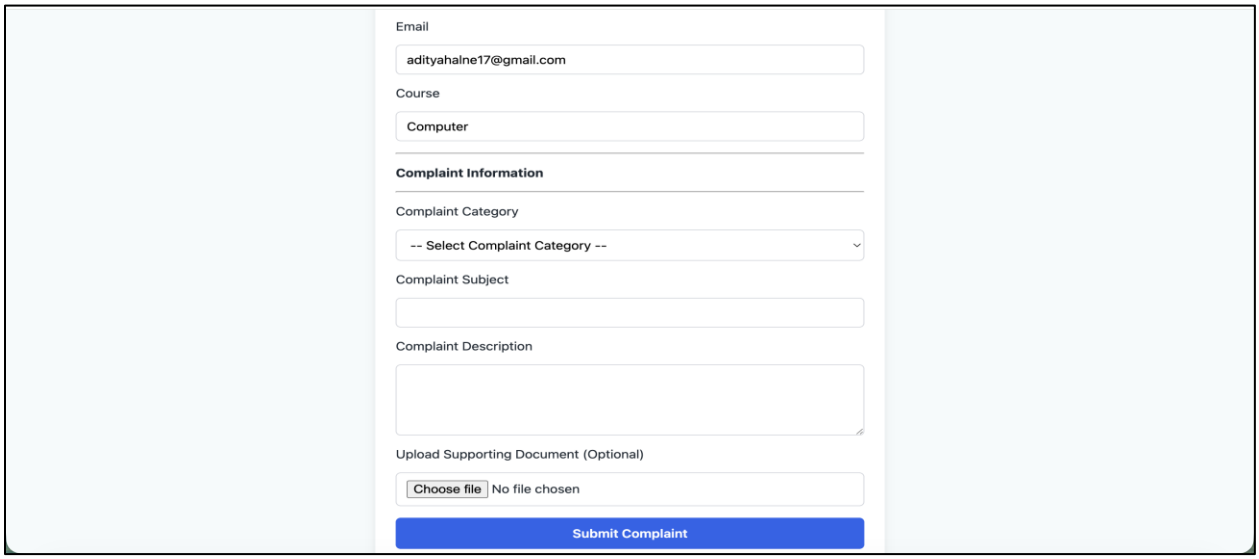


Screenshot No.5

6. Complaint Submission Form

Fields:

- Student Data
- Complaint Data



Screenshot No.6

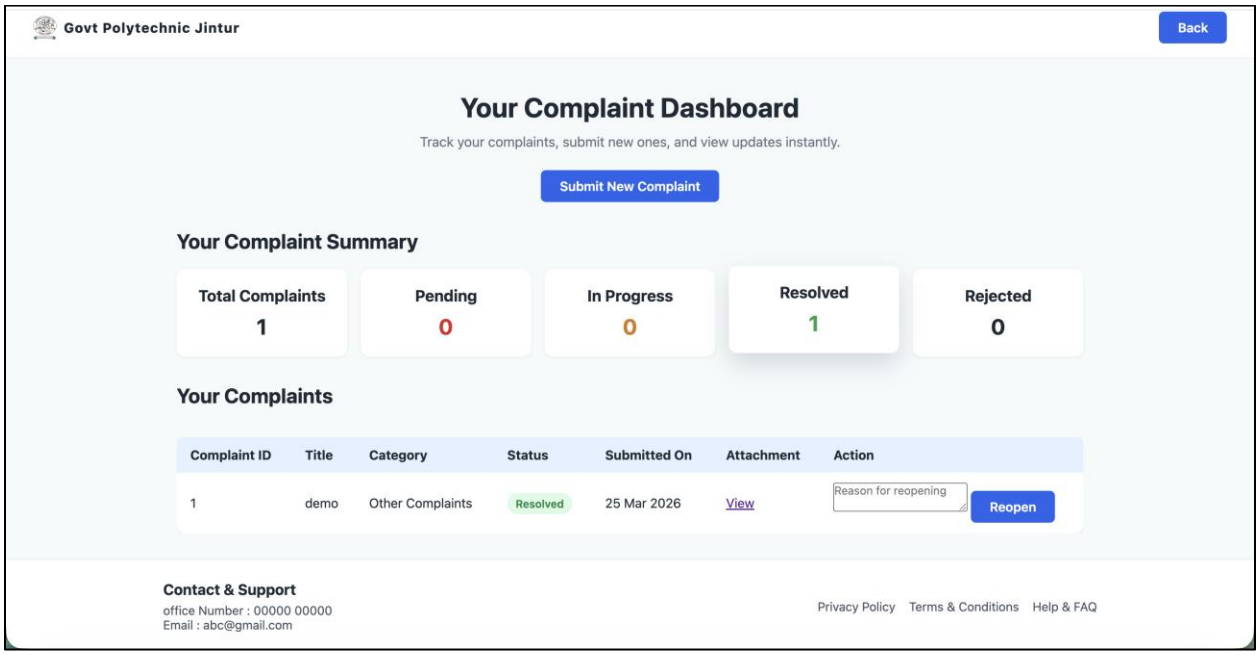
7. Complaint Tracking Page

Description:

Table format:

- Complaint ID

- Subject
- Status
- Date
- Option to reopen complaint

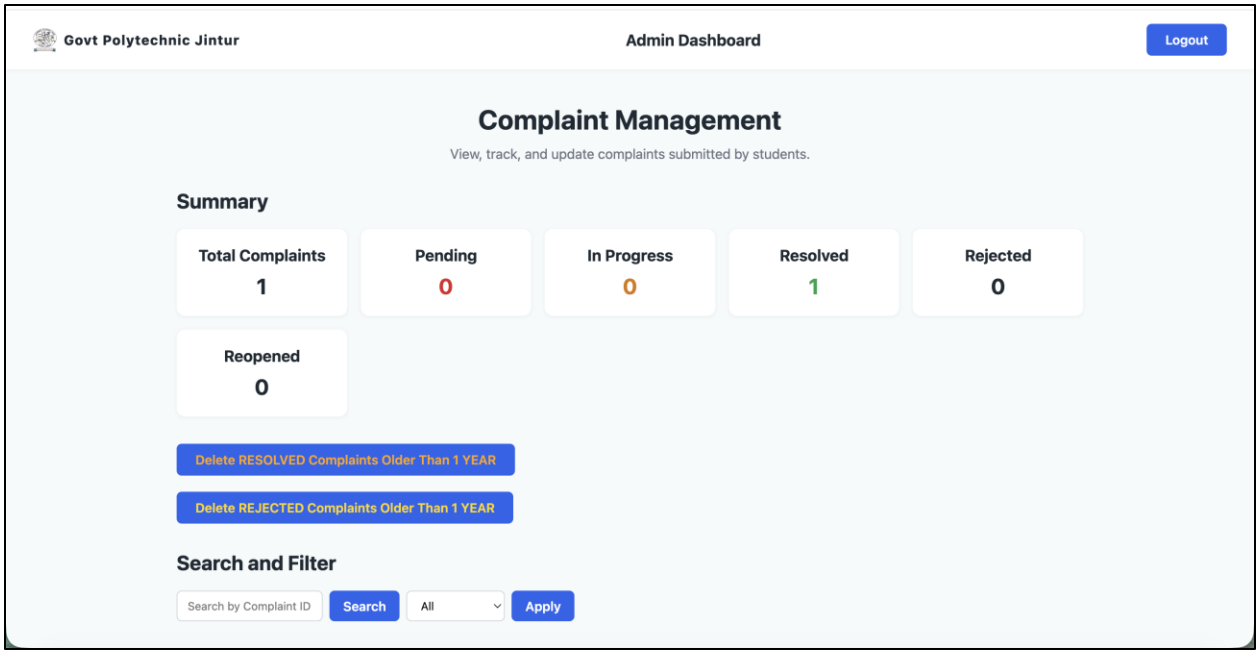


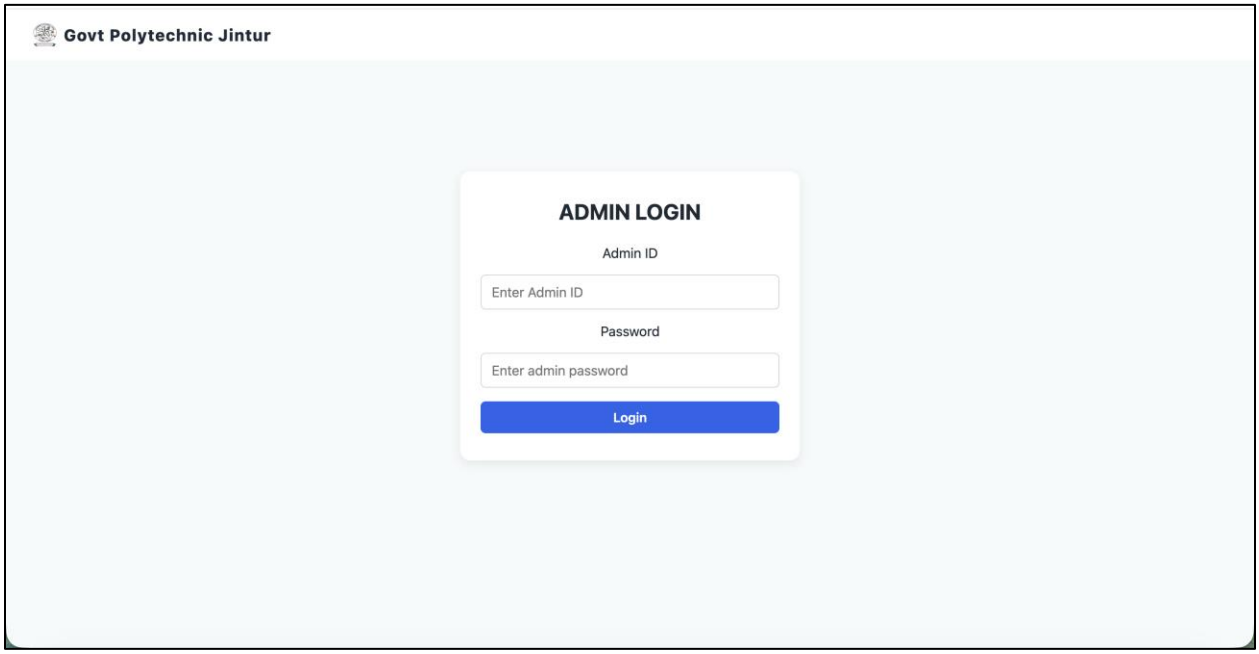
Screenshot No.7

8. Admin Dashboard

Description:

- Displays overall complaint statistics
- Includes filter options



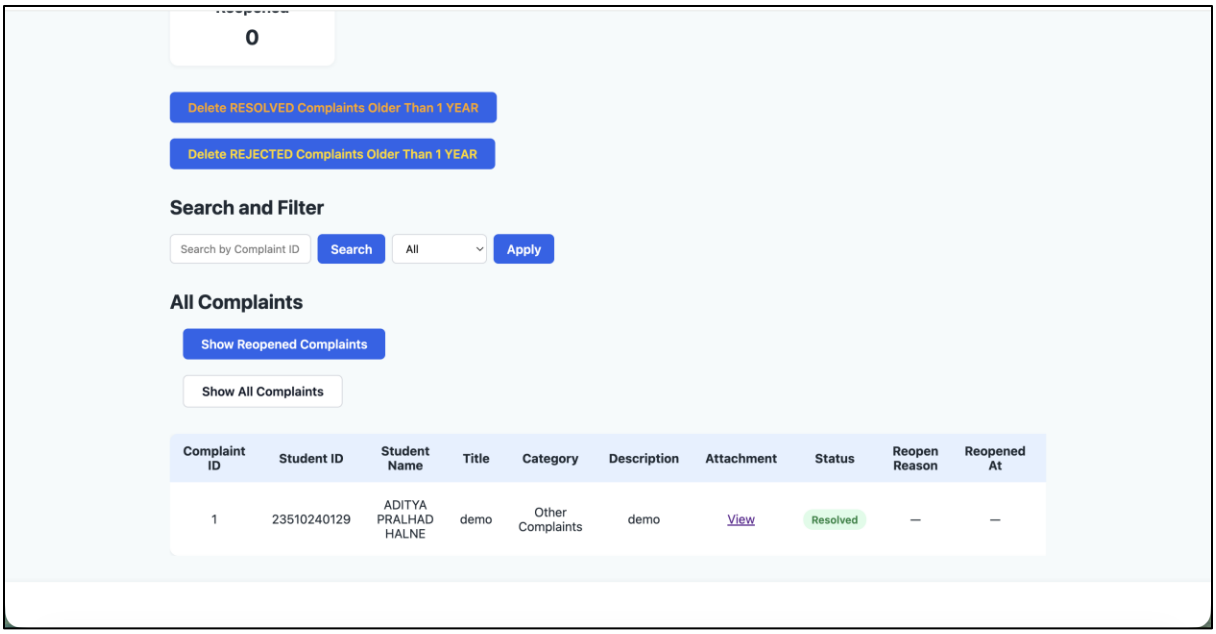


Screenshot No.8 & 9

9. Admin Complaint Table

Description:

- Displays all complaints
- Includes:
- Student details
- Complaint info
- Status update option



Screenshot No.10

8.4 Discussion

The system performs efficiently and meets all objectives:

- Easy to use
- Secure authentication
- Proper complaint tracking
- Effective admin control

8.5 Conclusion

The results show that the system is fully functional and improves grievance handling efficiency.

CHAPTER 9: CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The Grievance Redressal Portal has been successfully developed and implemented. It provides an efficient, transparent, and secure platform for handling student complaints.

The system overcomes the limitations of manual processes by:

- Providing real-time complaint tracking
- Ensuring secure user authentication
- Allowing structured complaint management
- Improving communication between students and administration

Overall, the project meets all the defined objectives and requirements.

9.2 Achievements of the System

1. Successful implementation of OTP-based authentication
2. Efficient complaint submission system
3. Real-time tracking of complaint status
4. Admin control for complaint management
5. Secure and reliable system

9.3 Limitations

1. Limited to web platform
2. No multi-admin support
3. Requires internet connection
4. Basic UI design

9.4 Future Scope

The system can be enhanced with the following features:

1. Mobile Application

Develop Android/iOS application for better accessibility.

2. AI-Based Chatbot

Integrate chatbot to assist students in filing complaints.

3. SMS Notifications

Send updates via SMS in addition to email.

4. Advanced Analytics

Provide reports and graphs for complaint analysis.

5. Multi-Admin System

Allow multiple administrators with role-based access.

6. Integration with ERP

Integrate with college management systems.

9.5 Conclusion

The project demonstrates how technology can improve administrative processes and enhance student satisfaction.

❑ REFERENCES

1. Online Documentation

1. PHP Official Documentation
- Used for backend scripting and server-side logic
- Covers syntax, functions, sessions, and form handling
2. MySQL Documentation
- Used for database design and queries

- Covers table creation, joins, constraints

3. PHPMailer Documentation

- Used for sending OTP emails
- Covers SMTP configuration and email handling

4. XAMPP Documentation

- Used for local server setup
- Covers Apache and MySQL configuration

2. Research Papers and Articles

- Research papers on grievance management systems
- Articles on web-based complaint handling
- Online tutorials for PHP and MySQL integration

3. Websites and Tutorials

W3Schools (HTML, CSS, PHP basics)

Stack Overflow (problem-solving and debugging)

YouTube tutorials (project implementation guidance)

CHAPTER 10: APPENDICES

10.1 Introduction

Appendices include additional information such as code snippets, screenshots, and supporting materials that are not included in the main chapters but are important for understanding the system.

10.2 Appendix A: Sample Code

1. Database Connection Code (PHP):

```
$conn = mysqli_connect("localhost", "root", "", "grievance_portal");

if (!$conn) {

   die("Connection failed: " . mysqli_connect_error()); }
```

2. OTP Generation Code:

```
$otp = rand(100000, 999999);
```

3. Send OTP using PHPMailer:

```
$mail->setFrom('official_email@gmail.com');

$mail->addAddress($user_email);

$mail->Subject = "OTP Verification";

$mail->Body = "Your OTP is: " . $otp;

$mail->send ();
```

4. Complaint Submission Code:

```
$query = "INSERT INTO complaints (student_id, subject, category, description)

VALUES ('$student_id', '$subject', '$category', '$description')";

mysqli_query($conn, $query);
```

10.3 Appendix A: Database Structure

Include screenshots of:

Tables in phpMyAdmin

Table structure (fields and types)

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	**full_name**	varchar(100)	utf8mb4_general_ci		No	*None*			Change Drop More
3	**student_id**	varchar(50)	utf8mb4_general_ci		No	*None*			Change Drop More
4	**email**	varchar(100)	utf8mb4_general_ci		No	*None*			Change Drop More
5	**course**	varchar(50)	utf8mb4_general_ci		No	*None*			Change Drop More
6	**complaint_category**	varchar(100)	utf8mb4_general_ci		No	*None*			Change Drop More
7	**title**	varchar(255)	utf8mb4_general_ci		No	*None*			Change Drop More
8	**description**	text	utf8mb4_general_ci		No	*None*			Change Drop More
9	**status**	varchar(20)	utf8mb4_general_ci		Yes	Pending			Change Drop More
10	**submitted_at**	timestamp			No	current_timestamp()			Change Drop More
11	**attachment**	varchar(255)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
12	**reopen_reason**	text	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
13	**reopened_at**	datetime			Yes	*NULL*			Change Drop More

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**srno**	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	**name**	varchar(30)	utf8mb4_general_ci		No	*None*			Change Drop More
3	**student_id**	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More
4	**email**	varchar(225)	utf8mb4_general_ci		No	*None*			Change Drop More
5	**password**	varchar(225)	utf8mb4_general_ci		No	*None*			Change Drop More
6	**date**	date			No	current_timestamp()			Change Drop More
7	**course**	varchar(30)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More

Table No. 5

10.4 Appendix B: Test Data

Example:

Student ID: 23510240129

Email: adityahalne17@gmail.com

Password: Adi123!@#

10.5 Conclusion

Appendices provide supporting details that help in better understanding the implementation of the system.

CHAPTER 11: SOURCE CODE LINK AND EXECUTION INSTRUCTIONS

11.1 Introduction

This chapter provides the source code link and step-by-step instructions to run the project successfully.

11.2 GitHub Repository Link

https://github.com/adityaph-17/grievance-redressal-portal

11.3 Steps to Run the Project

Step 1: Install XAMPP

- Download and install XAMPP server
- Start Apache and MySQL

Step 2: Setup Project Folder

- Copy project folder
- Paste into: C:\xampp\htdocs\ or xampp\htdocs(for macOS or Linux)

Step 3: Import Database

- Open phpMyAdmin
- Create database: grievance_portal
- Import .sql file

Step 4: Configure Database Connection

- Open PHP file
- Update: $conn = mysqli_connect("localhost", "root", "", "grievance_portal");

Step 5: Configure Email (PHPMailer)

- Enable SMTP
- Use Gmail App Password
- Update email credentials in code

Step 6: Run the Project

- Open browser and type:
- http://localhost/project-folder (path of your project folder)

11.4 Important Notes

1. Ensure internet connection for email OTP
2. Check spam folder for OTP emails
3. Use correct database credentials

11.5 Conclusion

Following the above steps ensures successful execution of the project.

....THANK-YOU....